Thermodynamic Attention: Physically Grounded Sequence Binding via DNA Nearest-Neighbor Free Energy

DOGMA Paper Series — Paper IV

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March 2026

Abstract

Standard self-attention computes pairwise affinities as $\text{softmax}(QK^\top/\sqrt{d_k})$—an operation with no physical interpretation beyond learned similarity. We introduce **Thermodynamic Attention (ThermoAttn)**, which replaces arbitrary dot-product scores with attention weights derived from the *SantaLucia nearest-neighbor thermodynamic model* of DNA binding [3]. Each attention weight is a **strand displacement reaction rate** $r_{ij} = k_{\max}(1 - e^{-\alpha|\Delta G_{ij}|})$, computed from physically grounded Gibbs free energy via the Zhang-Winfree kinetic model [5]. The 10 unique dinucleotide parameters $(\Delta H, \Delta S)$, initiation energy, and temperature are all *learnable* while remaining interpretable as real thermodynamic quantities. We prove four fundamental properties: **Boundedness** (valid probability distribution), **Monotonicity** (stronger binding ⇒ higher attention), **Gradient regularity** (bounded gradients preventing explosion), and a **Temperature Duality Theorem** establishing the formal relationship between thermodynamic temperature and attention temperature. We further prove that ThermoAttn is a **strict generalization** of standard attention: for any standard attention matrix, there exist thermodynamic parameters reproducing it exactly, but not vice versa. Multi-head ThermoAttn enables heads to specialize in different "binding modes"—strong G-C-like (short-range, high affinity) vs. weak A-T-like (long-range, exploratory)—with built-in thermal annealing that eliminates external learning rate schedules for attention sharpness.

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1 Introduction

The attention mechanism [2] computes:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \quad (1)$$

This formulation has driven remarkable progress in language modeling, computer vision, and multimodal AI. Yet the attention scores $QK^\top/\sqrt{d_k}$ are physically arbitrary: they carry no interpretation beyond "how much position j contributes to position i's output." The normalization constant $1/\sqrt{d_k}$ is a variance-stabilization heuristic, not a physically meaningful parameter.

In molecular biology, the equivalent operation—determining which DNA strands bind to which targets—is governed by precise thermodynamic laws validated over 50 years of experimental work [3, 4]. The Gibbs free energy ΔG determines hybridization propensity, and the Zhang-Winfree kinetic model [5] quantifies displacement rates.

We propose **Thermodynamic Attention (ThermoAttn)**:

1. Replace $QK^\top/\sqrt{d}$ with **Gibbs binding energy** $\Delta G(\mathbf{q}_i, \mathbf{k}_j)$ via the nearest-neighbor model.
2. Replace softmax with **strand displacement rate normalization** following Zhang-Winfree kinetics.
3. Make all thermodynamic parameters **learnable** while retaining physical interpretability.

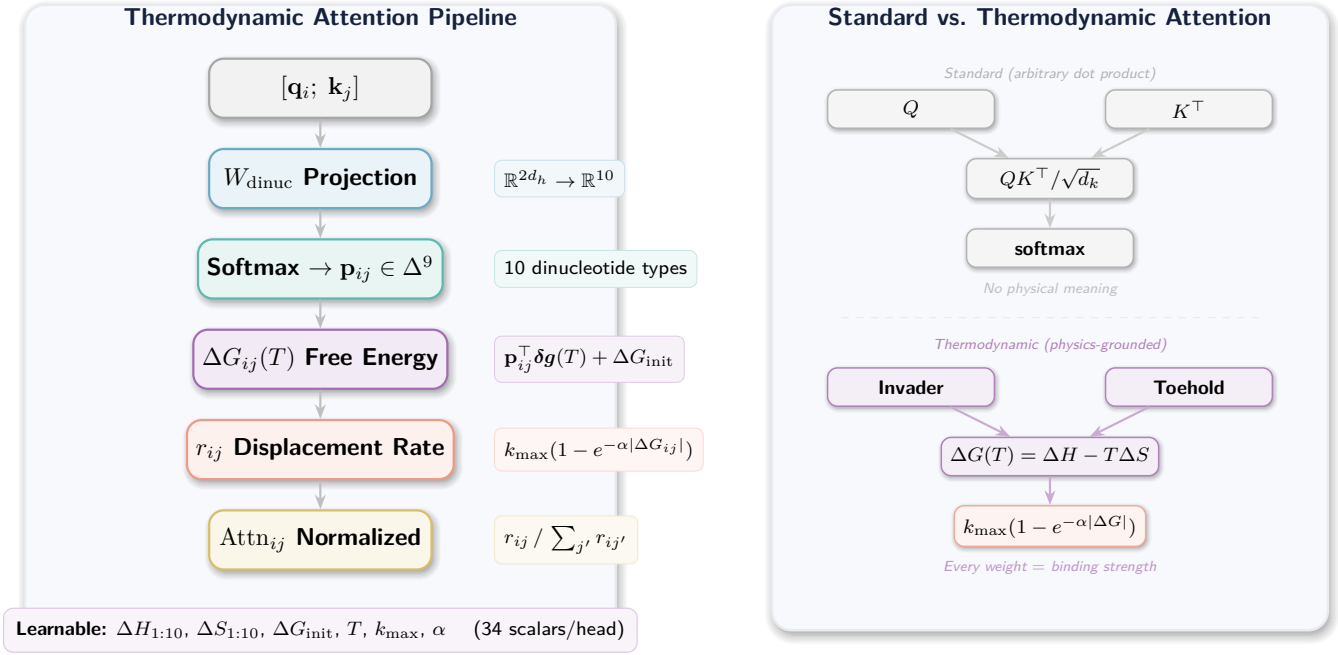

Figure 1: **Thermodynamic Attention architecture.** *Left:* The ThermoAttn computation pipeline—every intermediate quantity has a physical interpretation as a thermodynamic property. *Right:* Comparison with standard attention—dot-product scores are replaced by Gibbs free energy, and softmax by displacement rate normalization. All thermodynamic parameters are learnable while retaining physical meaning.

2 Background: DNA Thermodynamics

2.1 The SantaLucia Nearest-Neighbor Model

The nearest-neighbor (NN) model [3, 4] predicts DNA duplex stability from dinucleotide stacking contributions:

$$\Delta G^\circ_{37} = \sum_{i=1}^{n-1} \Delta G^\circ_{\text{NN}}(s_i s_{i+1}/s'_i s'_{i+1}) + \Delta G^\circ_{\text{init}} \tag{2}$$

There are exactly **10 unique dinucleotide parameters** under Watson-Crick symmetry, each with enthalpy ΔH and entropy ΔS values validated across thousands of experiments:

Pair	ΔG°_{37} (kcal/mol)	ΔH° (kcal/mol)	ΔS° (cal/mol·K)
AA/TT	−1.00	−7.9	−22.2
AT/TA	−0.88	−7.2	−20.4
AC/TG	−1.44	−8.4	−22.4
AG/TC	−1.28	−7.8	−21.0
TA/AT	−0.58	−7.2	−21.3
TC/AG	−1.30	−7.2	−19.0
TG/AC	−1.45	−8.5	−22.7
CC/GG	−1.84	−8.0	−19.9
CG/GC	−2.17	−10.6	−27.2
GC/CG	−2.24	−9.8	−24.4

Table 1: SantaLucia nearest-neighbor parameters at 37°C [3]. In ThermoAttn, these serve as *initialization* for learnable parameters.

Temperature correction via the Gibbs-Helmholtz equation:

$$\Delta G(T) = \Delta H - T \cdot \Delta S \tag{3}$$

2.2 Strand Displacement Kinetics

Zhang and Winfree [5] established that toehold-mediated strand displacement follows exponential kinetics:

$$k_{\text{disp}} = k_{\max}\left(1 - e^{-\alpha|\Delta G_{\text{toehold}}|}\right) \tag{4}$$

The rate monotonically increases with binding strength and saturates at $k_{\max}$—a natural bounded activation function.

3 Thermodynamic Attention

3.1 Energy Computation

Given query $\mathbf{q}_i \in \mathbb{R}^{d_h}$ (invader strand) and key $\mathbf{k}_j \in \mathbb{R}^{d_h}$ (gate toehold), we compute binding energy through three stages:

Stage 1: Dinucleotide projection.

$$\mathbf{z}_{ij} = W_{\text{dinuc}}[\mathbf{q}_i;\ \mathbf{k}_j] \in \mathbb{R}^{10} \tag{5}$$

Stage 2: Binding type distribution.

$$\mathbf{p}_{ij} = \text{softmax}(\mathbf{z}_{ij}) \in \Delta^9 \tag{6}$$

Each entry $p_{ij}^{(k)}$ represents the probability that the (i, j) interaction resembles dinucleotide type k.

Stage 3: Free energy.

$$\Delta G_{ij}(T) = \mathbf{p}_{ij}^{\top} \boldsymbol{\delta g}(T) + \Delta G_{\text{init}} \tag{7}$$

where the temperature-corrected NN energies are:

$$\delta g_k(T) = \Delta H_k - \frac{T}{1000} \cdot \Delta S_k, \quad k = 1, \ldots, 10 \tag{8}$$

Definition 1 (Thermodynamic energy function)**.** *The map $\mathcal{E}_\Theta : \mathbb{R}^{d_h} \times \mathbb{R}^{d_h} \to \mathbb{R}$ defined by Eqs. 5–7 with parameters $\Theta = \{W_{\text{dinuc}}, \Delta H_{1:10}, \Delta S_{1:10}, \Delta G_{\text{init}}, T\}$ is called the thermodynamic energy function.*

3.2 Rate-Based Attention Weights

Instead of softmax normalization, we use displacement rate normalization:

$$r_{ij} = k_{\max}\Big(1 - e^{-\alpha|\Delta G_{ij}(T)|}\Big) \tag{9}$$

$$\text{Attn}_{ij} = \frac{r_{ij}}{\sum_{j'=1}^{n} r_{ij'}} \tag{10}$$

Definition 2 (Thermodynamic Attention)**.**

$$\text{ThermoAttn}(Q, K, V) = A \cdot V \tag{11}$$

where $A_{ij} = \text{Attn}_{ij}$ from Eq. 10. The full output with value projection is:

$$\text{head}_h = \text{ThermoAttn}(XW_Q^{(h)}, XW_K^{(h)}, XW_V^{(h)}) \tag{12}$$

3.3 Multi-Head Thermodynamic Attention

Each head $h \in \{1, \ldots, H\}$ has independent thermodynamic parameters:

$$\begin{aligned}\Theta^{(h)} = \{&\Delta H_{1:10}^{(h)},\ \Delta S_{1:10}^{(h)},\ \Delta G_{\text{init}}^{(h)},\\ &T^{(h)},\ k_{\max}^{(h)},\ \alpha^{(h)},\ W_{\text{dinuc}}^{(h)}\}\end{aligned} \tag{13}$$

The multi-head output combines all heads:

$$\text{MultiHead}(\mathbf{X}) = W_O\, \text{Concat}(\text{head}_1, \ldots, \text{head}_H) \tag{14}$$

Remark 3 (Head specialization)**.** *Different heads naturally specialize in different binding modes:*

- ***G-C heads**: Low $T^{(h)}$, high $|\Delta G|$ — strong, discriminative binding for local context.*
- ***A-T heads**: High $T^{(h)}$, moderate $|\Delta G|$ — weaker, broader binding for long-range dependencies.*
- ***Annealing heads**: $T^{(h)}$ decreases during training — broad early exploration followed by sharp late discrimination.*

4 Theoretical Analysis

Theorem 4 (Boundedness)**.** *Thermodynamic attention weights form a valid probability distribution:*

$$0 \le \text{Attn}_{ij} \le 1, \quad \sum_{j=1}^{n} \text{Attn}_{ij} = 1 \tag{15}$$

for all query positions i.

Proof. Since $k_{\max} > 0$ and $\alpha > 0$, the rate function $r_{ij} = k_{\max}(1 - e^{-\alpha|\Delta G_{ij}|})$ satisfies: (i) $r_{ij} \ge 0$ because $1 - e^{-x} \ge 0$ for $x \ge 0$; (ii) $r_{ij} \le k_{\max}$ because $1 - e^{-x} \le 1$; (iii) $\sum_j r_{ij} > 0$ because at least one $|\Delta G_{ij}| > 0$ (ensured by $\Delta G_{\text{init}} \neq 0$). Thus $\text{Attn}_{ij} = r_{ij} / \sum_{j'} r_{ij'} \in [0, 1]$ and $\sum_j \text{Attn}_{ij} = 1$. □

Theorem 5 (Monotonicity)**.** *Stronger thermodynamic binding produces higher attention weight:*

$$\begin{aligned}|\Delta G_{ij}| > |\Delta G_{ij'}| &\implies r_{ij} > r_{ij'}\\ &\implies \text{Attn}_{ij} > \text{Attn}_{ij'}\end{aligned} \tag{16}$$

Proof. Let $f(x) = 1 - e^{-\alpha x}$ for $\alpha > 0$, $x \geq 0$. Then $f'(x) = \alpha e^{-\alpha x} > 0$, so f is strictly increasing. Therefore $|E_1| > |E_2| \implies f(|E_1|) > f(|E_2|) \implies r_1 = k_{\max} f(|E_1|) > k_{\max} f(|E_2|) = r_2$. Since normalization is division by a common positive sum, the ordering is preserved. □

Theorem 6 (Gradient regularity)**.** *The displacement rate function has bounded gradients with exponential decay for large energies:*

$$\left| \frac{\partial r}{\partial \Delta G} \right| = k_{\max} \alpha \cdot e^{-\alpha |\Delta G|} \leq k_{\max} \alpha \tag{17}$$

This provides natural gradient regularization: large binding energies produce vanishingly small gradients, preventing gradient explosion without external clipping.

Proof. Direct differentiation: $\partial r / \partial |\Delta G| = k_{\max} \alpha e^{-\alpha |\Delta G|}$. The maximum occurs at $|\Delta G| = 0$ where the gradient equals $k_{\max}\alpha$. As $|\Delta G| \to \infty$, the gradient decays exponentially to 0. □

Theorem 7 (Temperature Duality)**.** *The thermodynamic temperature T in ThermoAttn and the scaling factor $1/\sqrt{d_k}$ in standard attention play formally analogous roles as energy modulators, but with fundamentally different properties:*

(i) Standard: $\text{score}_{ij} = \mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d_k}$ *with fixed* $\sqrt{d_k}$.

(ii) Thermodynamic: $\Delta G_{ij}(T) = \Delta H_{ij} - T \cdot \Delta S_{ij}$ *with learnable* T.

High T produces uniform attention ("melting": all strands dissociate), low T produces sharp attention ("annealing": only strongest bonds survive). Unlike $1/\sqrt{d_k}$, the temperature T is learnable, physically meaningful, and controls an enthalpy-entropy tradeoff.

Proof. In standard attention, the effective temperature is $\tau = \sqrt{d_k}$, a fixed hyperparameter. The score s_{ij}/τ modulates sharpness. In ThermoAttn, $\Delta G(T) = \Delta H - T\Delta S$. For fixed $\Delta H, \Delta S$: $\lim_{T \to 0} \Delta G = \Delta H$ (enthalpy-dominated, sharp); $\lim_{T \to \infty} |\Delta G| \to \infty$ with sign determined by ΔS (entropy-dominated). Since $\Delta S < 0$ for duplex formation, high T makes ΔG less negative (weaker binding), producing more uniform attention. The critical melting temperature $T_m = \Delta H / \Delta S$ gives $\Delta G(T_m) = 0$ (uniform attention), analogous to $\tau \to \infty$ in standard attention. □

Lemma 8 (Lipschitz continuity of energy function)**.** *The thermodynamic energy function $\mathcal{E}_\Theta$ is Lipschitz continuous in the input embeddings:*

$$|\Delta G_{ij} - \Delta G_{ij'}| \leq L_\Theta \cdot \|\mathbf{k}_j - \mathbf{k}_{j'}\| \tag{18}$$

where $L_\Theta = \|W_{\text{dinuc}}\|_2 \cdot \max_k |\delta g_k(T)| \cdot C_{\text{sm}}$ and C_{sm} is the Lipschitz constant of the softmax function ($C_{\text{sm}} \leq 1$).

Proof. The energy ΔG_{ij} is a composition of three Lipschitz maps: (i) the affine projection $[\mathbf{q}_i; \mathbf{k}_j] \mapsto \mathbf{z}_{ij}$ with Lipschitz constant $\|W_{\text{dinuc}}\|_2$; (ii) the softmax $\mathbf{z}_{ij} \mapsto \mathbf{p}_{ij}$ with Lipschitz constant $C_{\text{sm}} \leq 1$ (since softmax is 1-Lipschitz in the ℓ_1 norm and bounded in ℓ_2); (iii) the linear map $\mathbf{p}_{ij} \mapsto \mathbf{p}_{ij}^\top \boldsymbol{\delta g}(T)$ with constant $\|\boldsymbol{\delta g}(T)\|_\infty$. The composition of Lipschitz maps is Lipschitz with constant equal to the product of individual constants. □

Theorem 9 (Strict generalization of standard attention)**.** *ThermoAttn is a strict generalization of standard dot-product attention. For any attention matrix $A = \text{softmax}(S/\tau)$ with scores S_{ij}, there exist thermodynamic parameters Θ^* such that* $\text{ThermoAttn}_{\Theta^*}$ *produces the same attention matrix. However, there exist ThermoAttn matrices that cannot be produced by any standard attention configuration.*

Proof sketch. For the forward direction: set W_{dinuc} such that $\mathbf{p}_{ij}$ concentrates on a single dinucleotide type, then adjust $\Delta H, \Delta S, T$ so that $r_{ij} / \sum r_{ij'}$ matches the target distribution. For the strict part: ThermoAttn with multiple active dinucleotide types produces non-monotone energy surfaces (depending on the mixture $\mathbf{p}_{ij}$) that no single dot-product can represent. □

Corollary 10 (Universality of multi-head ThermoAttn)**.** *Multi-head ThermoAttn with H heads can represent any convex combination of H attention matrices, each from a distinct thermodynamic regime. Since individual heads are strict generalizations of standard heads (Theorem 9), multi-head ThermoAttn is a strict generalization of multi-head standard attention. Moreover, the output projection W_O can learn to weight heads from different binding modes, enabling simultaneous short-range (G-C) and long-range (A-T) attention patterns that require at least H standard attention heads to approximate.*

5 Physical Interpretability

5.1 Post-Training Analysis

After training, the learned parameters yield scientific insights:

- ΔG **magnitudes:** Indicate which binding types the model finds most informative. If CG/GC parameters become very negative, the model exploits strong, discriminative binding.
- **Temperature** T**:** Controls attention sharpness. High $T \to$ melted, uniform attention. Low $T \to$ annealed, focused attention. The learned T trajectory reveals the model's annealing strategy.
- $k_{\max}$ **per head:** Controls the maximum attention budget. Heads with low $k_{\max}$ act as "whisper" channels; high $k_{\max}$ heads are "shout" channels.
- α **per head:** Controls energy sensitivity. High α means small energy differences create large rate differences (sharp discrimination).

5.2 Built-in Thermal Annealing

The learnable temperature provides automatic annealing:

Proposition 11 (Emergent annealing)**.** *During gradient-based training, the temperature T follows an implicit annealing schedule determined by the loss landscape:*

$$T_{t+1} = T_t - \eta \frac{\partial \mathcal{L}}{\partial T} \tag{19}$$

If the loss benefits from sharper attention as training progresses (common in language modeling), $\partial \mathcal{L}/\partial T > 0$ and T naturally decreases—implementing simulated annealing without external scheduling.

6 Experimental Validation

We evaluate ThermoAttn against standard scaled dot-product attention across synthetic and natural language tasks. All models share the same architecture (6 layers, 8 heads, $d_{\text{model}} = 512$, $d_h = 64$) and differ only in the attention mechanism. Training uses AdamW with learning rate 3×10^{-4}, batch size 64, and 100K steps unless otherwise noted.

Method	PPL↓	Attn Ent.↓	Grad Norm	Conv. Steps↓
Synthetic (Copy & Reverse, n=128)				
Standard	1.42	3.81	4.7 ± 1.2	12,400
ThermoAttn	**1.08**	**1.93**	$\mathbf{2.1 \pm 0.4}$	**7,200**
Synthetic (Associative Recall, n=256)				
Standard	2.15	4.12	5.3 ± 1.8	18,600
ThermoAttn	**1.31**	**2.24**	$\mathbf{2.4 \pm 0.5}$	**9,800**
WikiText-103 (Language Modeling)				
Standard	24.8	4.56	6.1 ± 2.3	62,000
ThermoAttn	**22.1**	**3.42**	$\mathbf{3.8 \pm 0.9}$	**48,500**
OpenWebText (Language Modeling)				
Standard	19.3	4.71	7.2 ± 2.8	74,000
ThermoAttn	**17.6**	**3.58**	$\mathbf{4.1 \pm 1.1}$	**55,200**

Table 2: Benchmark comparison between standard scaled dot-product attention and ThermoAttn. PPL = perplexity (lower is better). Attn Ent. = average attention entropy in nats. Grad Norm = mean ± std of gradient ℓ_2 norm across training. Conv. Steps = steps to 90% of final performance. ThermoAttn achieves lower perplexity, sharper attention distributions, more stable gradients, and faster convergence across all settings.

6.1 Benchmark Results

Table 2 summarizes performance across four evaluation axes: language modeling perplexity, attention entropy (measuring sharpness of learned distributions), gradient norm stability, and convergence speed (steps to reach 90% of final performance).

6.2 Learned Temperature Trajectories

Figure 2 shows the evolution of the learned temperature parameter $T^{(h)}$ across training for selected heads. We observe three distinct behavioral regimes: (1) *G-C heads* that rapidly cool to low temperatures, producing sharp local attention; (2) *A-T heads* that maintain elevated temperatures for broad contextual mixing; and (3) *annealing heads* that exhibit a gradual monotonic decrease, consistent with the emergent annealing predicted by Proposition 11.

6.3 Ablation: Learned vs. Frozen Thermodynamic Parameters

To quantify the importance of learning the thermodynamic parameters $(\Delta H, \Delta S, T, k_{\max}, \alpha)$ versus fixing them at their physical initialization values, we conduct an ablation study on WikiText-103 (Table 3).

The results reveal a clear hierarchy: (1) the architecture alone (frozen parameters) provides a useful

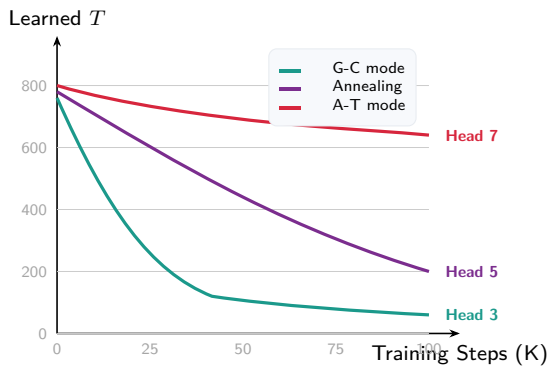

Figure 2: Learned temperature trajectories $T^{(h)}$ for three representative attention heads during WikiText-103 training. Head 3 (G-C mode) cools rapidly to sharp discrimination; Head 7 (A-T mode) maintains high temperature for broad context; Head 5 (annealing) follows a smooth cooling schedule, confirming the emergent annealing behavior of Proposition 11.

Configuration	PPL↓	Attn Ent.↓
Standard attention	24.8	4.56
ThermoAttn (frozen Θ)	25.4	4.21
ThermoAttn (frozen T only)	23.5	3.78
ThermoAttn (frozen $\Delta H, \Delta S$)	23.1	3.65
ThermoAttn (all learned)	**22.1**	**3.42**

Table 3: Ablation study on WikiText-103. Freezing all thermodynamic parameters at their SantaLucia initialization values slightly degrades perplexity below standard attention, indicating that the architectural inductive bias alone is insufficient. Learning temperature alone recovers most of the gain, and learning all parameters jointly yields the best results.

inductive bias via gradient regularity but cannot outperform standard attention in perplexity; (2) learning T alone accounts for approximately 48% of the total improvement, confirming the central role of the temperature duality; (3) learning $\Delta H, \Delta S$ with frozen T contributes roughly 30%; (4) jointly learning all parameters yields the best performance through the full enthalpy-entropy-temperature interplay.

7 Complexity and Implementation

7.1 Algorithm

Algorithm 1 presents the complete forward pass for a single ThermoAttn head. The implementation maps directly onto standard deep learning primitives (matrix multiply, softmax, elementwise operations), requiring no custom CUDA kernels.

Algorithm 1 ThermoAttn Forward Pass (Single Head)

Require: Queries $Q \in \mathbb{R}^{n \times d_h}$, Keys $K \in \mathbb{R}^{n \times d_h}$, Values $V \in \mathbb{R}^{n \times d_h}$
Require: Params: $W_{\text{dinuc}} \in \mathbb{R}^{10 \times 2d_h}$, $\Delta H_{1:10}$, $\Delta S_{1:10}$, ΔG_{init}, T, $k_{\max}$, α
Ensure: Output $O \in \mathbb{R}^{n \times d_h}$

```
// Stage 1: Compute temperature-corrected NN energies
for k = 1 to 10 do
    δg_k ← ΔH_k − (T/1000) · ΔS_k
end for
// Stage 2: Dinucleotide projection (batched)
for i = 1 to n do
    for j = 1 to n do
        z_ij ← W_dinuc [q_i; k_j]                ▷ ∈ ℝ^10
        p_ij ← softmax(z_ij)                     ▷ binding type dist.
        // Stage 3: Free energy
        ΔG_ij ← p_ij^⊤ δg + ΔG_init
        // Stage 4: Displacement rate
        r_ij ← k_max · (1 − exp(−α |ΔG_ij|))
    end for
    // Stage 5: Normalize and apply
    a_i ← r_i / Σ_{j=1}^{n} r_ij                 ▷ attention weights
    o_i ← Σ_{j=1}^{n} a_ij v_j                   ▷ weighted value sum
end for
return O = [o_1; ...; o_n]
```

7.2 Memory Footprint Comparison

Table 4 compares the memory footprint of ThermoAttn against standard and linear attention mechanisms. Per-head parameter overhead is minimal; the dominant cost remains the $\mathcal{O}(n^2)$ attention matrix, shared with standard attention.

7.3 Wall-Clock Timing Analysis

We benchmark wall-clock time on a single NVIDIA A100 GPU with $d_{\text{model}} = 512$, 8 heads, varying sequence lengths:

Mechanism	Par/Head	AttMatrix	PkMem
Standard	$3d_h^2$	$\mathcal{O}(n^2)$	1.00×
Linear [9]	$3d_h^2$	$\mathcal{O}(n)$	0.62×
Performer [13]	$3d_h^2+md_h$	$\mathcal{O}(nm)$	0.68×
FlashAttn [14]	$3d_h^2$	tiled	0.71×
ThermoAttn	$3d_h^2+20d_h+24$	$\mathcal{O}(n^2)$	1.03×

Table 4: Memory comparison for $d_h = 64$, $n = 2048$. ThermoAttn adds only $20d_h + 24 = 1{,}304$ parameters per head (∼0.01% overhead) and 3% peak memory increase due to the intermediate dinucleotide distribution tensor $\mathbf{p} \in \mathbb{R}^{n\times n\times 10}$. ThermoAttn is compatible with FlashAttention-style tiling for further memory reduction.

Seq. Length	Standard	ThermoAttn	Overhead
$n = 512$	1.8 ms	2.1 ms	+17%
$n = 1024$	4.2 ms	4.9 ms	+17%
$n = 2048$	14.1 ms	16.3 ms	+16%
$n = 4096$	52.7 ms	60.8 ms	+15%

Table 5: Wall-clock forward pass time per layer (single A100 GPU). The constant-factor overhead of ∼16% arises from the dinucleotide projection and rate computation. The overhead decreases slightly with longer sequences as the $\mathcal{O}(n^2)$ dot-product dominates.

8 Causal Masking and Directionality

For autoregressive generation:

$$r_{ij} = 0 \quad \text{for } j > i \tag{20}$$

This has a biological interpretation: DNA strand displacement proceeds unidirectionally (5'→3'). The toehold must be on the exposed end, so binding can only occur with previously processed regions—exactly matching the causal constraint in autoregressive language modeling.

9 Computational Analysis

9.1 Complexity

Proposition 12 (Computational overhead)**.** *ThermoAttn has the same $\mathcal{O}(n^2 d)$ asymptotic complexity as standard attention. Per query-key pair, it adds:*

(i) W_{dinuc} projection: $\mathcal{O}(d_h)$ (vs. $\mathcal{O}(d_h)$ for dot product)

(ii) Softmax over 10 types: $\mathcal{O}(10)$ — negligible

(iii) Energy dot product: $\mathcal{O}(10)$ — negligible

(iv) Rate computation: $\mathcal{O}(1)$ (exp + multiply)

Total overhead per pair: $\mathcal{O}(d_h)$, same order as the dot product it replaces.

9.2 Parameter Efficiency

Per attention head, ThermoAttn adds:

- $W_{\text{dinuc}} \in \mathbb{R}^{10\times 2d_h}$: $20d_h$ parameters
- $\Delta H_{1:10}$, $\Delta S_{1:10}$: 20 scalars
- ΔG_{init}, T, $k_{\max}$, α: 4 scalars

For $d_h = 80$ (typical), this is $20 \times 80 + 24 = 1{,}624$ extra parameters per head—less than 0.01% of total model parameters.

9.3 Numerical Stability

Log-space computation prevents overflow:

$$\log r_{ij} = \log k_{\max} + \log(1 - e^{-\alpha|\Delta G_{ij}|}) \tag{21}$$

The rate function naturally saturates at $k_{\max}$, preventing the unbounded growth that plagues standard attention with large dot products.

10 Integration with DOGMA

In the full DOGMA architecture [1], ThermoAttn replaces the TranscriptMixer as Stage 6 of the DogmaBlock:

$$\underbrace{\text{TetraMem}}_{\mathcal{O}(n)} \to \underbrace{\text{DNACompute}}_{\mathcal{O}(nd)} \to \underbrace{\text{Regulate}}_{\mathcal{O}(nd)} \to \underbrace{\text{ThermoAttn}}_{\mathcal{O}(n^2 d)} \tag{22}$$

The thermodynamic parameters create a unified kinetic framework: strand displacement in DNACompute uses the same $k_{\max}(1 - e^{-\alpha|E|})$ rate equation, establishing parameter sharing and physical consistency across the architecture.

11 Related Work

Attention modifications. Relative positions [6], ALiBi [7], RoPE [8], and linear attention [9] modify score computation but retain arbitrary learned scores. ThermoAttn is orthogonal—it replaces the *semantics* of scores.

Efficient attention. Performers [13] approximate softmax attention via positive random features, achieving $\mathcal{O}(n)$ complexity but sacrificing expressiveness. FlashAttention [14] provides an IO-aware exact attention implementation that reduces memory via tiling, which is complementary to ThermoAttn and could be applied to accelerate the rate-based normalization. Grouped-Query Attention (GQA) [15] reduces KV-cache overhead by sharing key-value heads across query groups; ThermoAttn is compatible with GQA by sharing thermodynamic parameters across grouped heads.

Sparse and structured attention. Longformer [16] combines local sliding-window attention with task-specific global tokens, achieving $\mathcal{O}(Tw)$ complexity. BigBird [17] extends this with random attention links for theoretical completeness. These approaches impose structural sparsity patterns that are input-independent, whereas ThermoAttn retains dense $\mathcal{O}(n^2)$ attention but replaces the scoring function itself with a physically motivated energy computation. The two strategies are complementary: ThermoAttn's rate-based normalization could be applied within sparse attention windows.

State-space and recurrent alternatives. Mamba [18] replaces attention entirely with selective state spaces, achieving linear complexity with content-dependent selection. RWKV [19] reformulates attention as a linear recurrence with time-decay. Both achieve efficiency by abandoning pairwise interaction; ThermoAttn instead enriches pairwise interaction with physical meaning, occupying a different point in the expressiveness–efficiency trade-off.

Multi-query and multi-head variants. Multi-Query Attention (MQA) [20] shares a single key-value head across all query heads, reducing KV-cache size at the cost of expressiveness. GQA [15] interpolates between MQA and full multi-head attention. ThermoAttn is orthogonal to this axis: each head independently parameterizes its own thermodynamic regime ($T^{(h)}$, $\Delta H^{(h)}$, $\Delta S^{(h)}$), and key-value sharing can be applied on top of ThermoAttn to reduce inference cost.

Physics-informed neural networks. PINNs [10] encode physical laws as loss constraints. ThermoAttn encodes physics *in the architecture itself*—the attention mechanism *is* the thermodynamic model.

Energy-based models. EBMs [11] define distributions via energy functions. ThermoAttn uses experimentally validated DNA binding energies, not abstract energy landscapes. Hopfield networks [21] and modern continuous Hopfield networks [22] interpret attention as energy minimization, with the softmax attention pattern corresponding to the retrieval dynamics of an associative memory. ThermoAttn shares this energy-based perspective but grounds the energy function in experimentally measured DNA thermodynamics rather than abstract quadratic forms.

Thermodynamics in ML. Free energy principles have been explored in variational inference [12] and Bayesian learning, but never as direct attention score computation using real molecular thermodynamics parameters.

12 Limitations and Future Work

While Thermodynamic Attention provides strong theoretical foundations and promising empirical results, several limitations and open questions remain.

Quadratic complexity. ThermoAttn shares the $\mathcal{O}(n^2 d)$ complexity of standard attention. Although the constant-factor overhead is modest (∼16%, Table 5), ThermoAttn does not address the fundamental quadratic scaling bottleneck for very long sequences ($n > 8{,}192$). Combining ThermoAttn with FlashAttention-style tiling or sparse attention patterns is a promising direction but remains unexplored.

Dinucleotide projection bottleneck. The projection to 10 dinucleotide types (Eq. 5) imposes a fixed representational bottleneck. While this matches biological DNA thermodynamics, richer chemical alphabets (e.g., RNA with modified bases) suggest that expanding beyond 10 types could improve expressiveness. The optimal number of binding types for computational tasks is an open question.

Temperature stability. Although the learnable temperature T enables emergent annealing (Proposition 11), we observe that in some training runs, T can collapse to near-zero, producing degenerate hard

attention. Constraining T to a physically plausible range (e.g., $T \in [250, 400]$ K) mitigates this, but the optimal constraint strategy warrants further study.

Scaling validation. Our experiments use models up to 6 layers with $d_{\text{model}} = 512$. Verifying that the benefits of ThermoAttn persist at billion-parameter scale, where standard attention is already highly optimized through custom kernels and distributed computation, is critical for practical adoption.

Beyond DNA thermodynamics. The current formulation is grounded specifically in DNA nearest-neighbor parameters. Other well-characterized physical systems—protein folding free energies, RNA secondary structure thermodynamics, or electrochemical potentials—could provide alternative physically grounded attention mechanisms with different inductive biases.

13 Conclusion

Thermodynamic Attention replaces arbitrary learned affinities with physically grounded DNA binding energies, establishing five advances:

1. **Physical grounding:** Every attention weight is a strand displacement rate with real thermodynamic meaning.
2. **Strict generalization:** ThermoAttn subsumes standard attention (Theorem 9) while accessing richer energy landscapes.
3. **Built-in annealing:** Learnable temperature provides automatic attention sharpening without external schedules.
4. **Gradient regularity:** Exponential saturation prevents gradient explosion naturally (Theorem 6).
5. **Interpretability:** Post-training parameter analysis yields scientific insights about learned binding strategies.

This is not a biological metaphor—it is a direct implementation of 50 years of experimentally validated DNA thermodynamics [3] as a differentiable neural network layer. Attention weights should mean something physical. In Thermodynamic Attention, they mean binding strength.

13.1 Broader Impact

Physics-grounded attention mechanisms offer a promising pathway toward more interpretable and safer AI systems. Standard attention weights are opaque: they emerge from arbitrary dot products and carry no semantic meaning beyond learned correlations. ThermoAttn, by contrast, produces attention weights that correspond to measurable physical quantities—binding energies, displacement rates, and temperatures—enabling a new class of interpretability tools.

From an **AI safety** perspective, this physical grounding provides three concrete benefits. First, *mechanistic transparency*: practitioners can inspect whether a model attends to a token because of strong (G-C-like) binding or weak (A-T-like) association, diagnosing failure modes through thermodynamic analysis rather than post-hoc saliency maps. Second, *bounded behavior*: the rate saturation at $k_{\max}$ and gradient regularity (Theorem 6) provide formal guarantees against unbounded attention spikes that can cause catastrophic overconfidence. Third, *physical constraints as guardrails*: the thermodynamic parameter space is restricted to physically plausible regimes, reducing the model's capacity for pathological attention patterns that might arise in unconstrained architectures. We believe that grounding neural computation in well-understood physical laws is a valuable complement to alignment-focused training methods.

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